Lecture VI - Computational Kindness

Programming with Python

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Key Concepts

Topics from the Lecture

- Optimal Stopping: How to decide **when to stop looking** for better options.
- Explore/Exploit Tradeoff: **Balancing** between trying new things and sticking with known options.
- Caching: Storing and reusing information by **sticking to a task** to improve performance.
- Scheduling: Efficiently managing **tasks and time.**
- Randomness: Understanding and working with **uncertainty**.

Kindness

Computational Kindness

Question: An idea what that is?

- Idea of introducing **empathy in algorithms**
- **Consider human cognitive load and limitations**
- Making choices that **reduce mental burden** for others
- Creating systems that are **easy to understand and interact**

Santiago Ramón y Cajal employed the art of drawing to depict his observations and to convey scientific arguments. He understood the persuasive power of images, and his drawings helped convince other neuroscientists of his theories. The aesthetic appeal and subtle emotional qualities in Cajal's images underscore their scientific content. Today neuroscientists have access to vastly more complex visualization tools than Cajal did. Modern scientists use digital images, not drawings, to share their observations and put forth their hypotheses. As this section of the exhibition shows, their images convey a different kind of beauty. As with Cajal's drawings, these contemporary images were chosen by neuroscientists and art historians for both their scientific significance and their aesthetic impact.

Cajal focused on cellular structures that make up the nervous system, magnifying cells that were about the width of a human hair. Since the invention of the electron microscope in 1931, first used in neuroscience in the 1940s, brain cells can be examined at increasingly greater magnification levels. Through Magnetic Resonance Imaging (MRI) techniques, in use since the late 1970s, scientists can examine the entire human brain. Cajal was limited to dead tissue fixed on slides that he viewed through his ocular microscope; MRI technology allows scientists to visualize the living brain.

Scheduling Meetings

- **Propose specific times!**
- Don’t ask “when are you free?”
- Reduces cognitive load for others
- Transforms an open-ended problem

> ⚠ Now a simple yes/no decision!

Communication

- **Be explicit rather than implicit**
- Provide clear options instead
- Avoid open-ended questions
- State your preferences

> ⚠ Avoid deferring → “I’m fine with **whatever**”!

The Cost of Flexibility

- “I’m free whenever” **seems helpful**
- **But it is not!**

- Forces others to consider all possibilities
- Increases cognitive load
- Makes decision-making more complex

Principles

1. Reduce Options: Fewer choices lead to **better decisions**
2. Be Explicit: Clear constraints **help** others decide
3. Consider Cognitive Load: Design interactions minimizing **mental effort** for others
4. Make Decisions: **Taking responsibility** can be kind

Reflection

Think about it: How could you apply computational kindness in these scenarios?

- Planning a group dinner with friends
- Asking your professor for thesis feedback
- Coordinating a team project at work

How to continue?

How to continue learning?

- We have covered **a lot of topics**
- But there are **many more** to explore!

Bayes' Rule

- Fundamental theorem in probability theory
- Updates probability of hypothesis based on **new evidence**
- Used in **statistics, machine learning, and decision-making**

> ⚠ Helps make **informed predictions and decisions** under uncertainty in the real world!

Overfitting

- Occurs when a model **learns the training data too well**
- **Captures noise and outliers** rather than pattern
- Results in **poor generalization to new data**

> ⚠ Several ways to counter overfitting, e.g. cross-validation.

Game Theory

- Strategic interactions among **rational decision-makers**
- Analyzes situations with **multiple agents** and their strategies

> **Many applications** in economics, politics, and biology!

Networking

- Study of **information exchange** over networks
- Includes **understanding protocols and data transmission**
- Optimizes **network resources**

> **Important** in computer networks, the internet and social life!

Relaxation

- Simplifies **complex problems** to make them more tractable
- Involves **relaxing certain constraints**
- Provides insights or **approximate** solutions

Continue programming?

- The best way to continue learning is to **keep programming in the future**
- Potentially, you will continue to do so during your studies
- Coding in your Thesis is another great way to improve
- Try to find a way to apply programming in your work
- **There are many interesting topics to explore!**

Advent of Code

- Advent of Code is a fun way to keep programming
- Here you can solve programming puzzles during Advent
- It is completely free and ad-free and starts each year at 01.12.

That's it for the Lecture Series!

- We now have covered the basics of Python
- I hope you enjoyed the lecture and found it helpful
- If you have questions or feedback, **please let me know!**
- I wish you all the best for your studies and your career!

Q&A

Literature

Interesting literature to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[1]

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Here
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.



Think Python is a great book to start with. It's available online for free. Schrödinger Programmiert Python is a great alternative for German students, as it is a very playful introduction to programming with lots of examples.

More Literature

For more interesting literature, take a look at the literature list of this course.